

Internet of Things

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Outline

- ▶ Introduction
- ▶ Basic Edge Architecture
- ▶ Design of Edge computing
- ▶ Case studies of Edge
- ▶ Edge computing software development
- ▶ Application Area
- ▶ Future of Edge computing
- ▶ References

Introduction

- ▶ With the rapid development the number of smart devices connected to the Internet is increasing, resulting in large-scale data, which has caused problems such as bandwidth load, slow response speed, poor security, and poor privacy in traditional cloud computing models. Traditional cloud computing is no longer sufficient to support the diverse needs of today's intelligent society for data processing, so edge computing technologies have emerged. It is a new computing paradigm for performing calculations at the edge of the network. Unlike cloud computing, it emphasizes closer to the user and closer to the source of the data. At the edge of the network, it is light weight for local, small-scale data storage and processing.

Important area for edge computing



Industrial applications are transforming manufacturing.

Important area for edge computing



The Internet of Things (IoT) is transforming computing.

Important area for edge computing



Distance between sensors and data centers
makes computing difficult.

Important area for edge computing



Microsatellites are bringing IoT sensors into space.

Important area for edge computing



Agriculture is benefiting from big data analytics.

Solution for Realtime data



Edge Computing
Computing on the endpoint

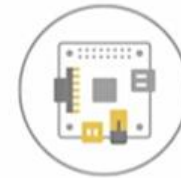
What are edge devices



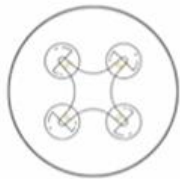
Mobile devices



Server in a data center



Small devices

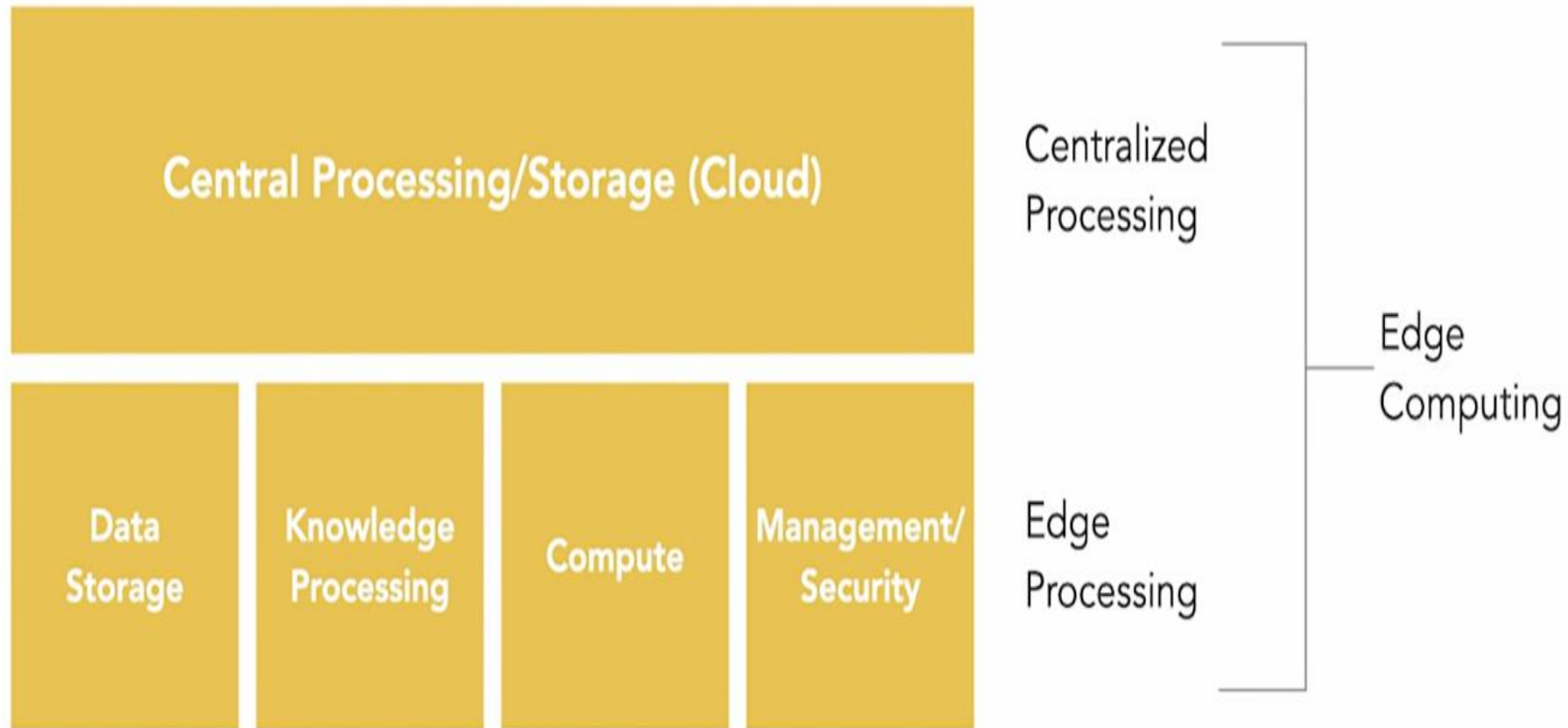


Drone



Medical devices

Basic Edge Architecture



Edge computing

- Network architectural pattern for compute and storage
- Also, centralized compute and storage

	Applicable situation	Network bandwidth pressure	Real-time	Calculation mode
Cloud computing	Global	More	High	Large scale centralized processing
Edge computing	Local	Less	Low	Small scale intelligent analysis

What edge device support

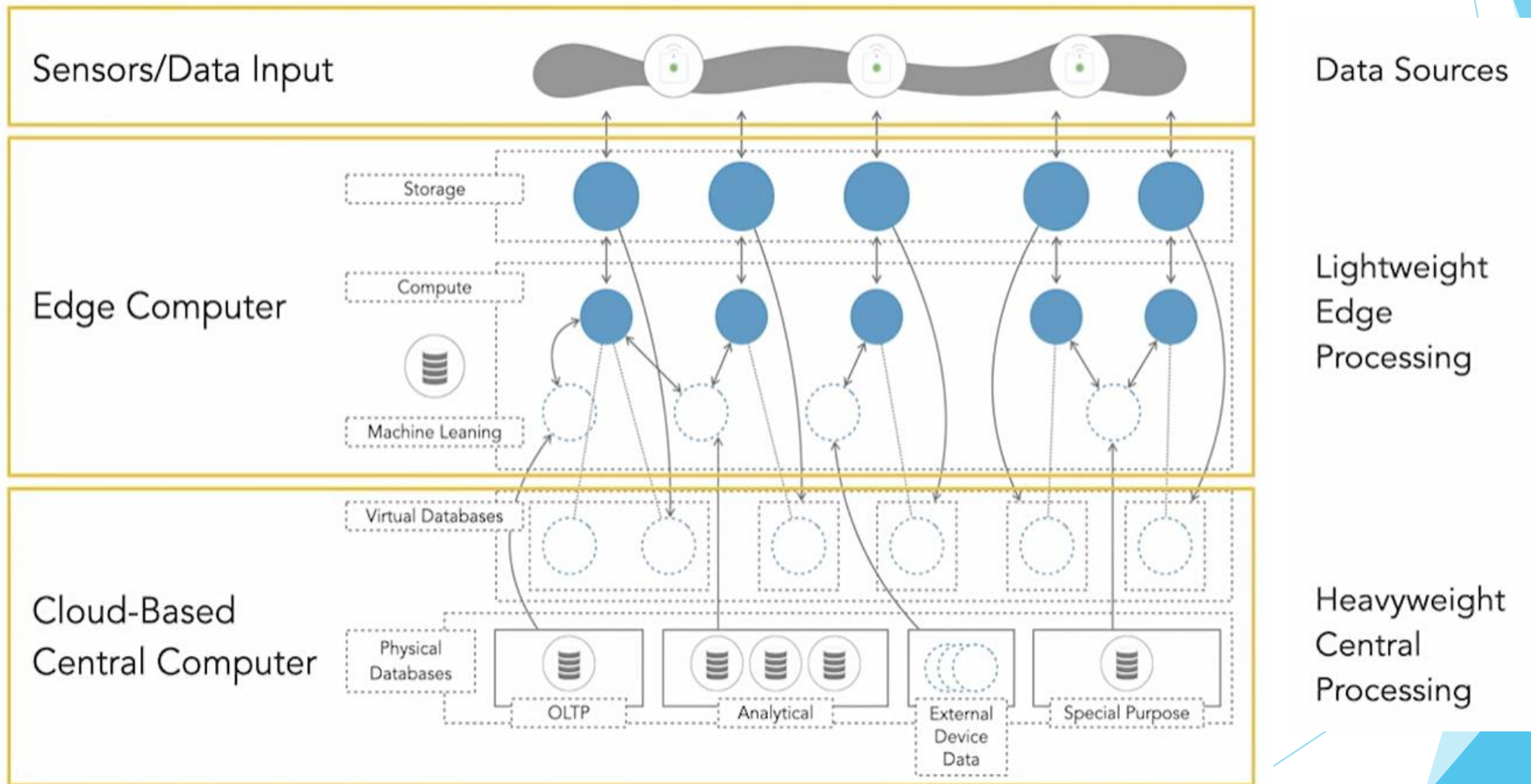
- Compute
- Storage
- Networking
- Self-contained
- Located to best support the system

Example of edge device

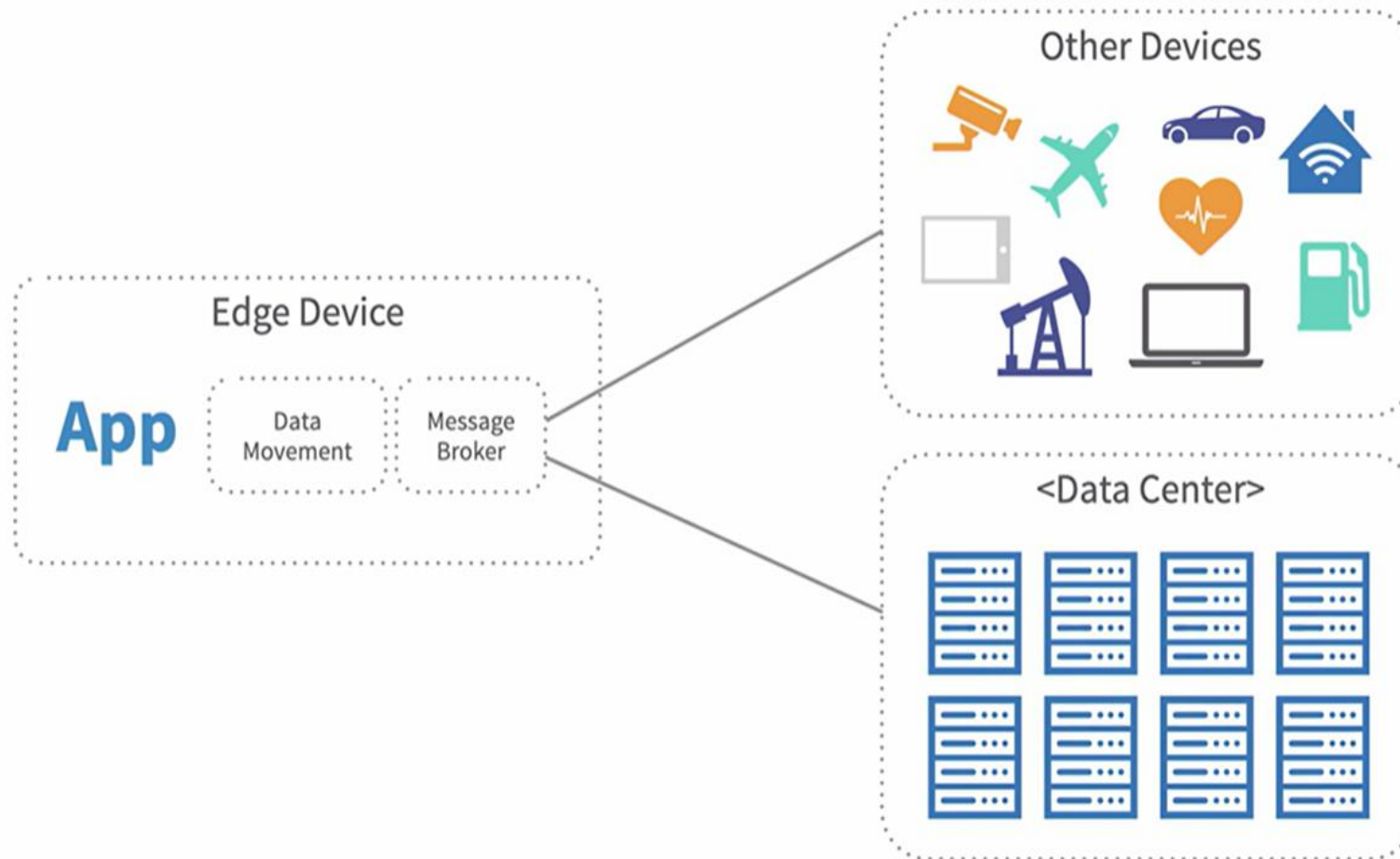
- **Raspberry Pi**
 - Storage
 - CPU
 - Networking



Basic Edge Architecture

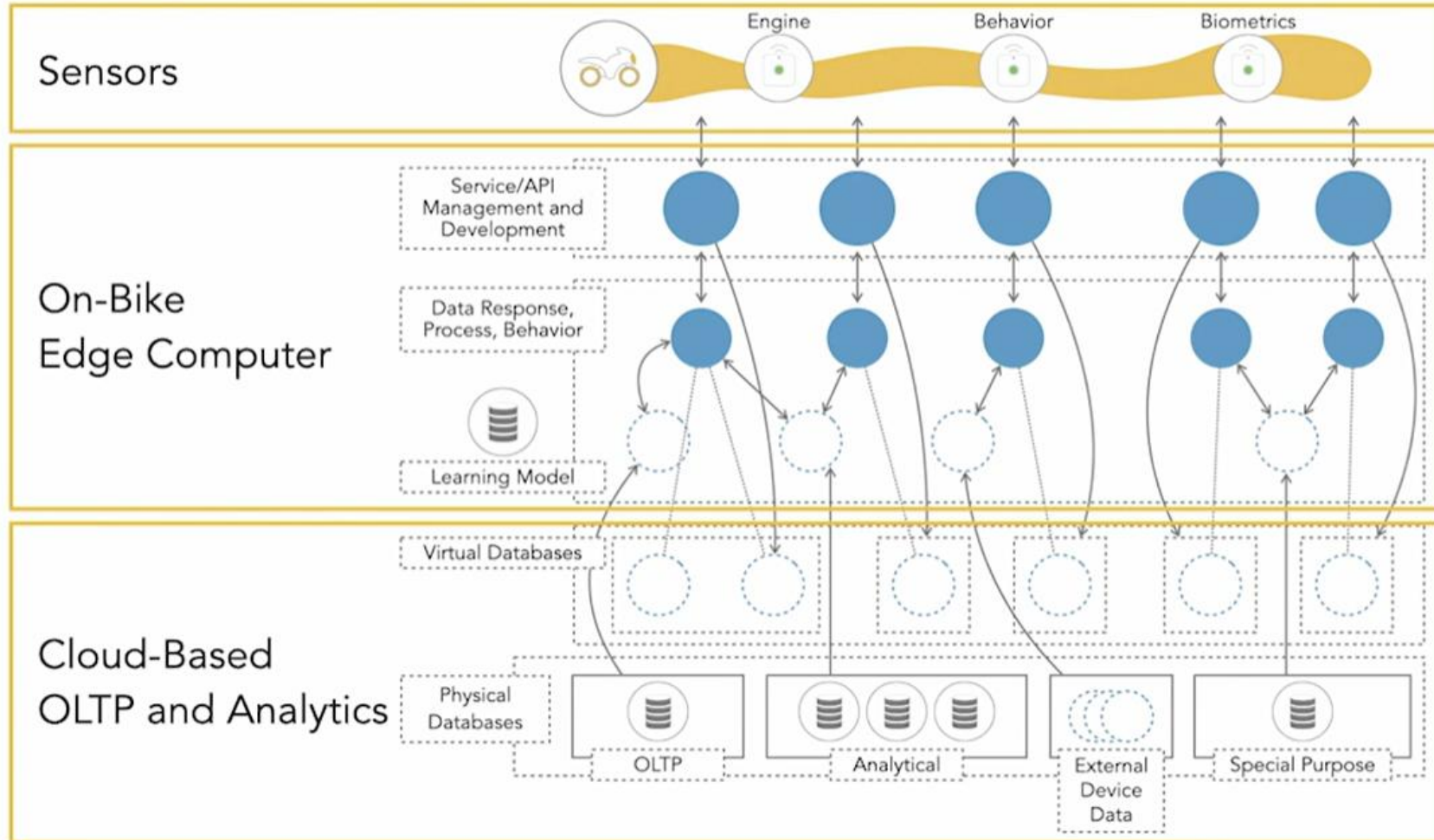


Basic Edge Architecture

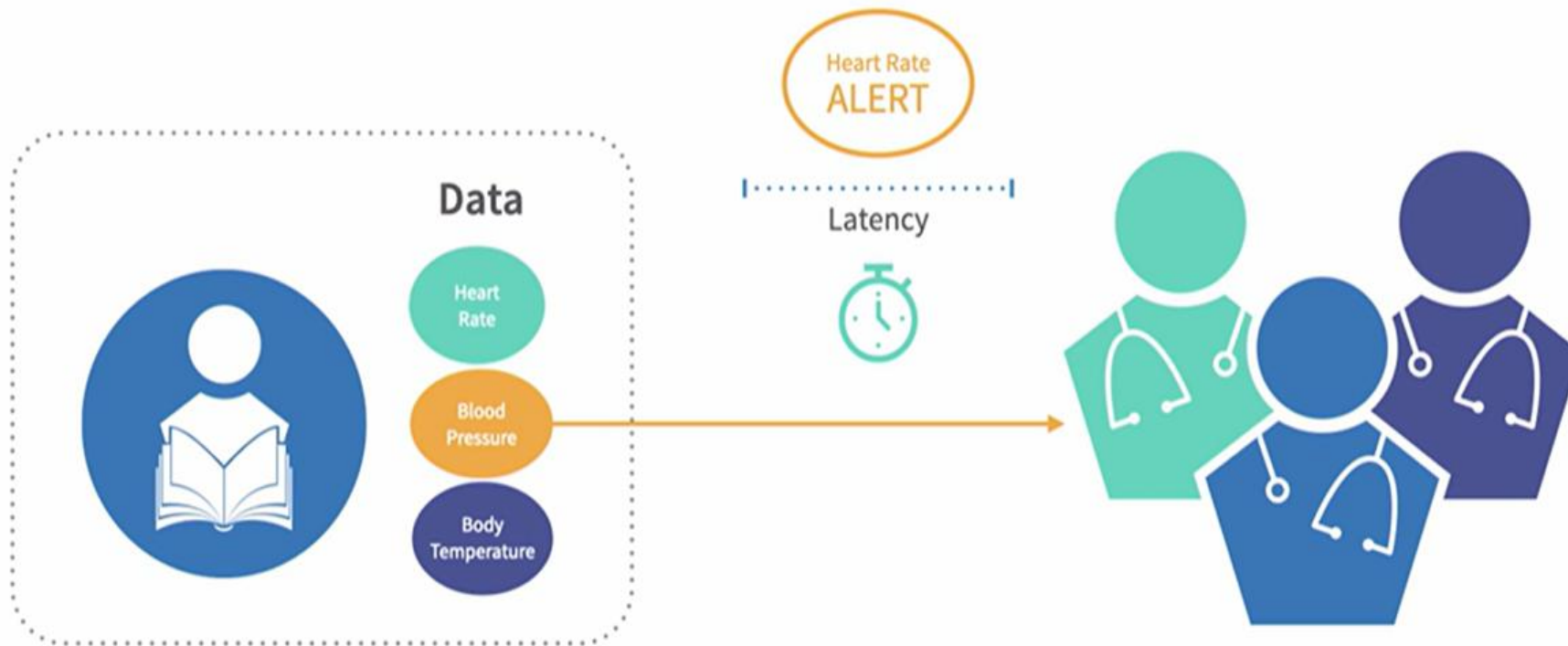


Edge use case

Connected Motorcycle



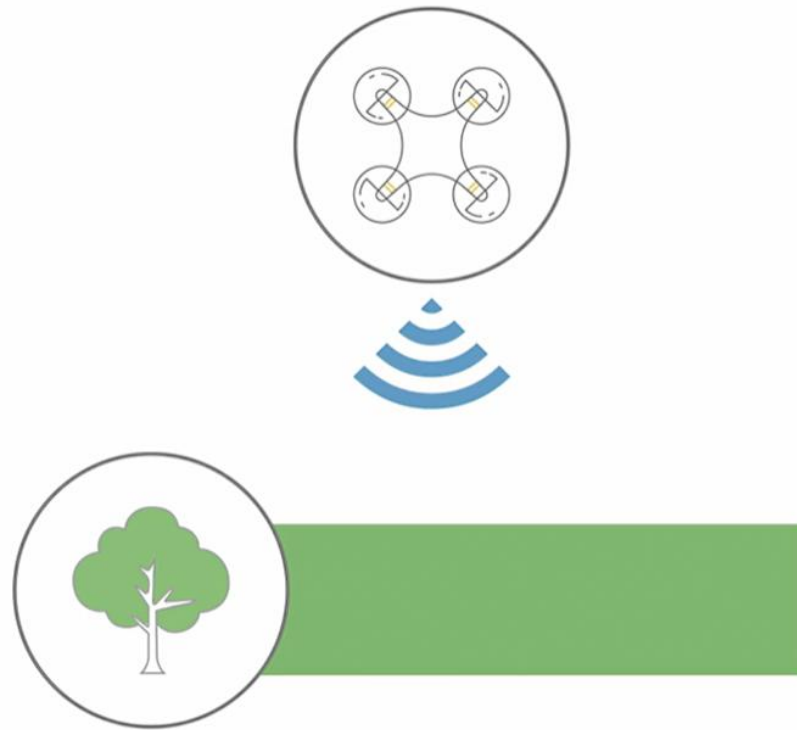
Edge use case: Patient information



Edge use case

Use Case: Saving Water

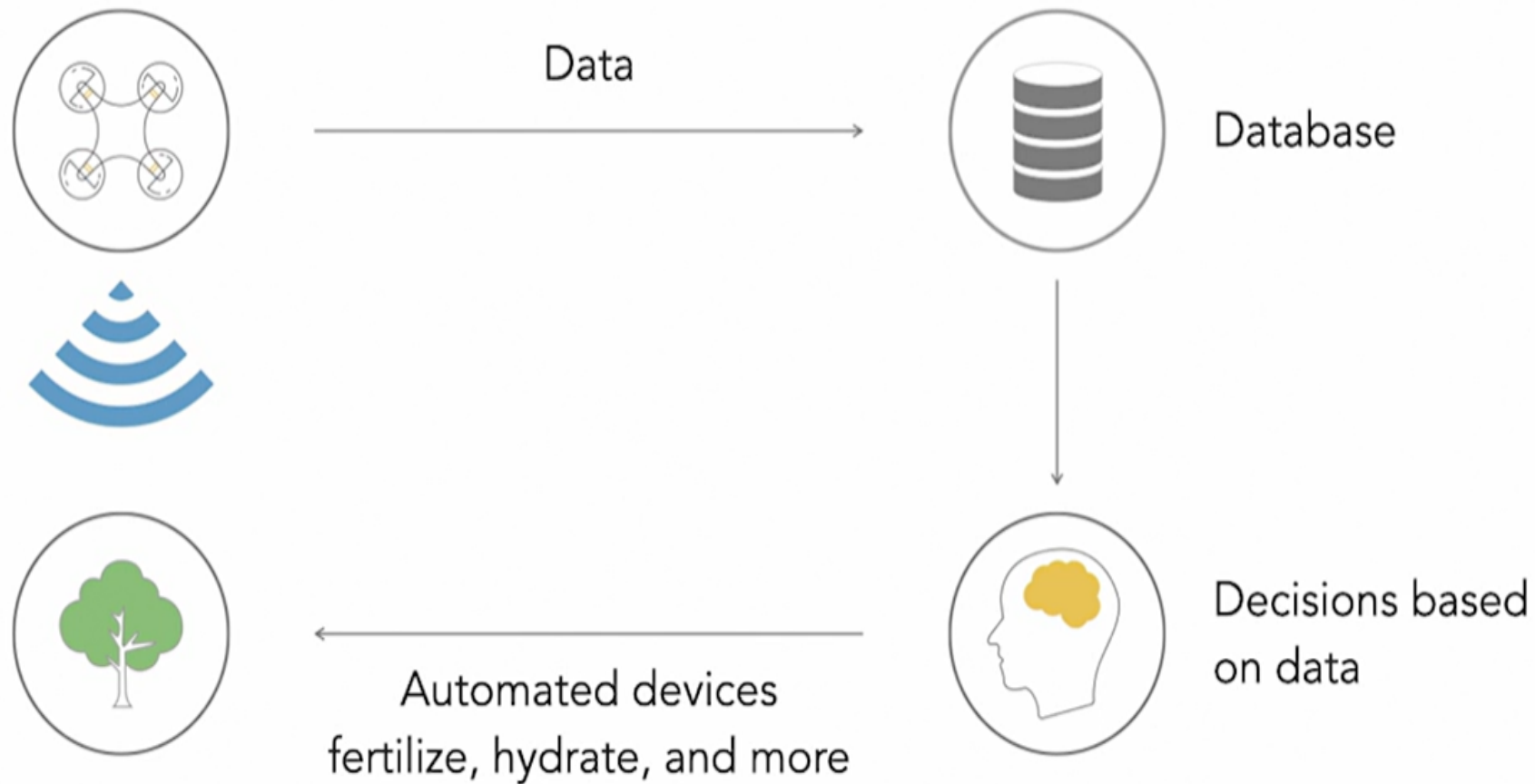
- **Collect data:**
 - Hydration
 - Weed saturation



Energy

- ▶ Many energy production facilities are in remote or hostile environments such as desert regions for solar arrays, steep hillsides for wind farms, and hazardous facilities for nuclear reactors. Additionally, data may need real-time or near real-time response for critical response to energy production control systems

Use case : Saving water



Components of Edge computing

Sensors

Sensors/Data Input

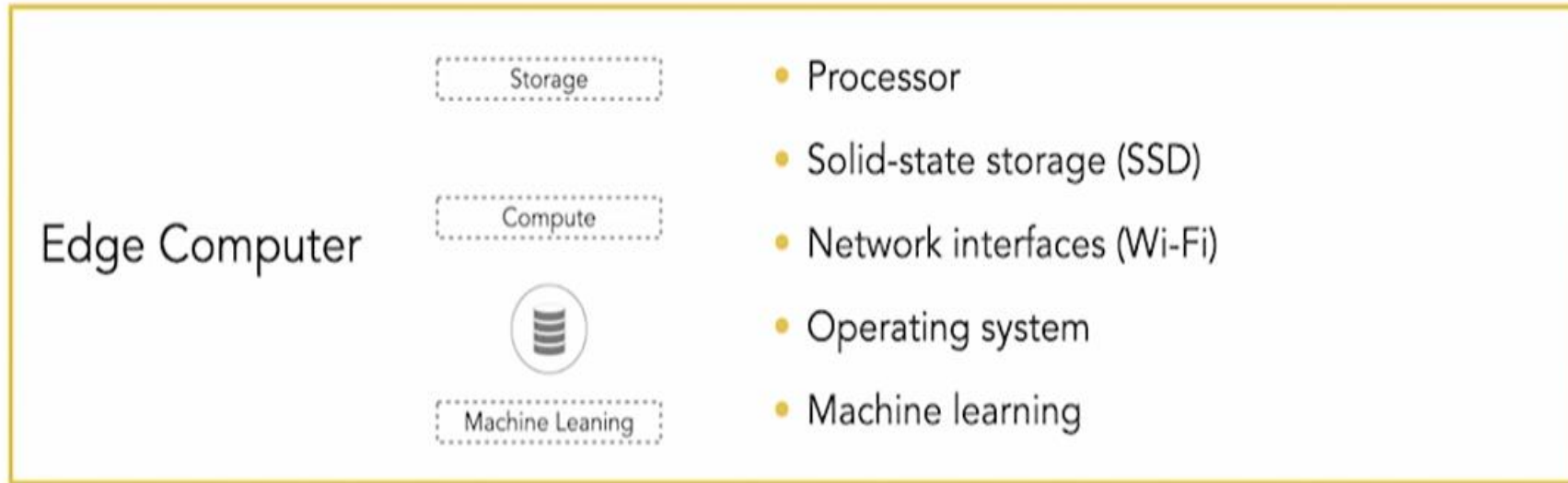


- Temperature
- Humidity
- Location



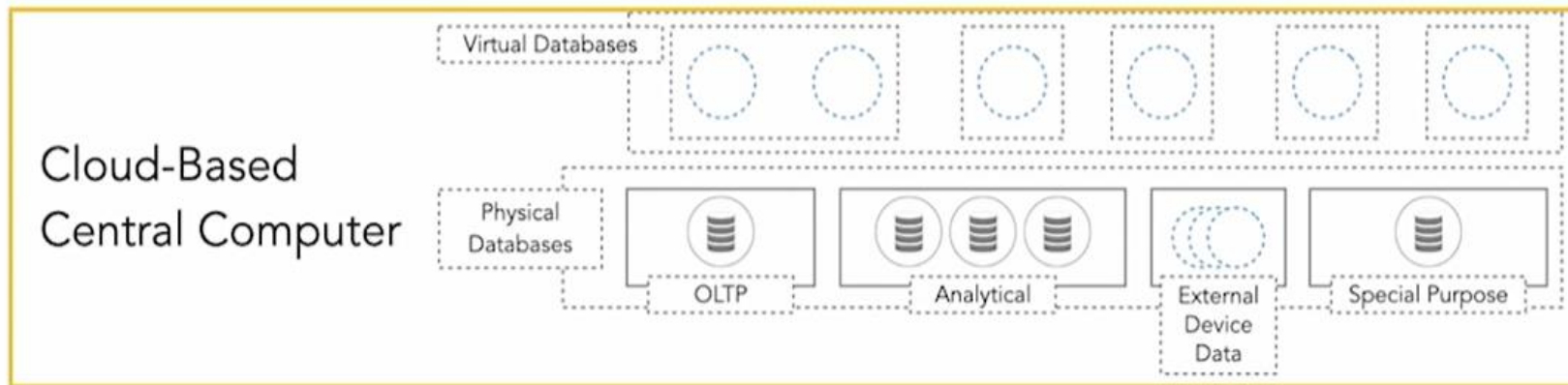
- Accident detection
- Reverse detection

Edge devices or computer



Cloud based central computer

- Virtual processor
- Mass storage
- Network interfaces
- Virtual operating systems
- Machine learning and analytics
- Edge system management



AWS IoT

- Connect and manage your devices
- Act on device data
- Communicate and set device state

Edge computing infrastructure

► Sensor consideration

- Standards
- Security
- Sensor management



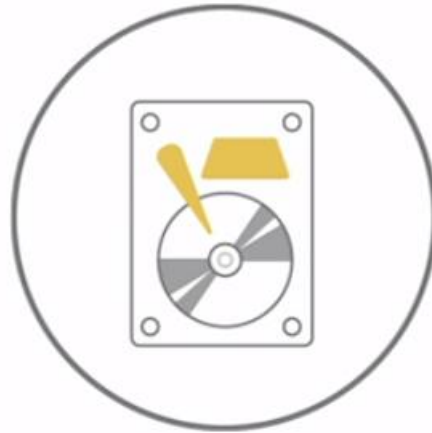
Network and wireless consideration

- 5G and cellular networks
- Bluetooth, and other wireless standards
- Network management
- Network security



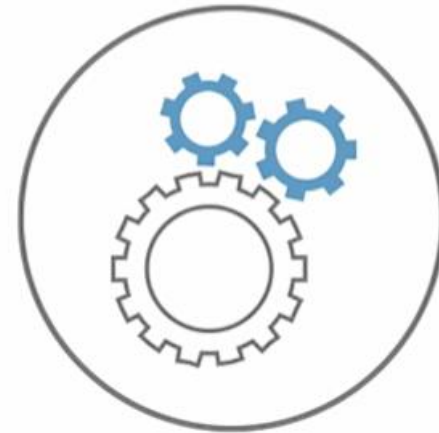
Storage consideration

- Physical storage
- Data security
- Data management

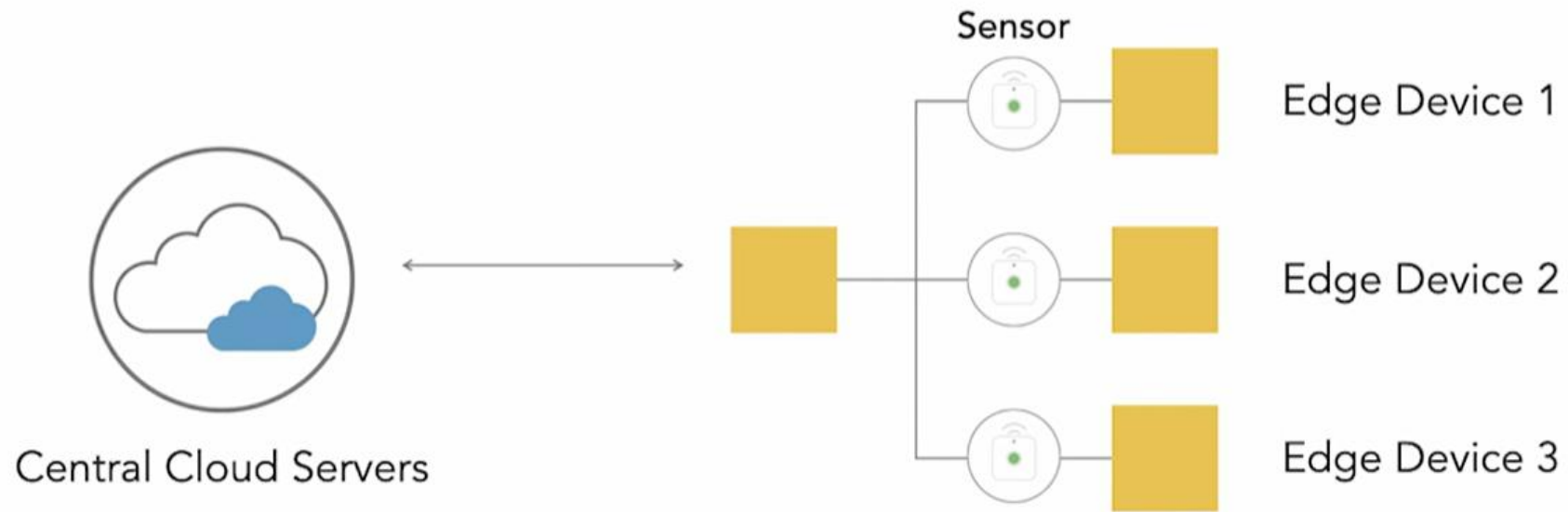


Configuration management

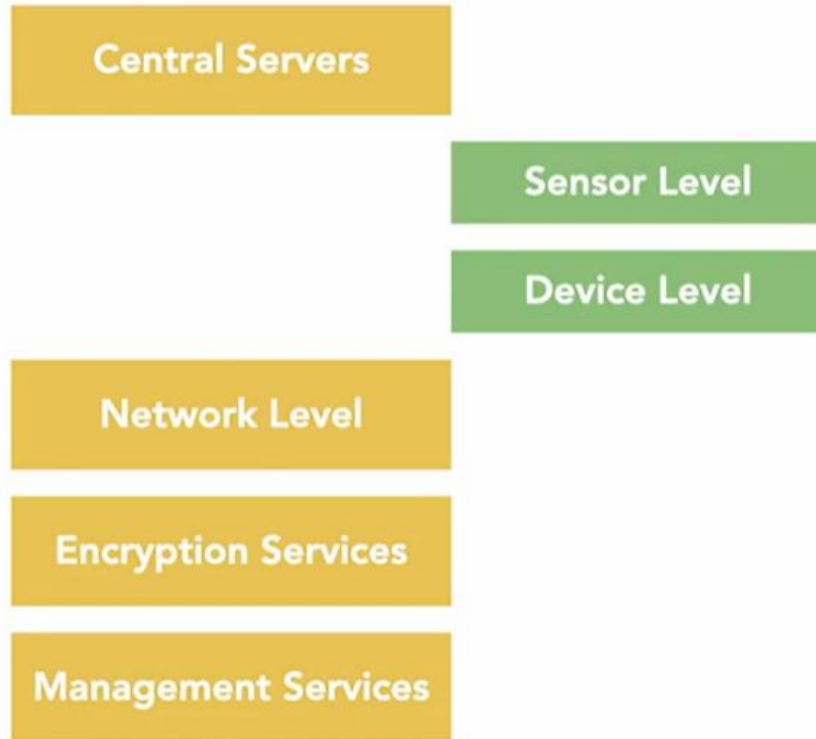
- Upgrade and fixes
- Remote device management
- Hard drive updates and upgrades



Edge computing networking



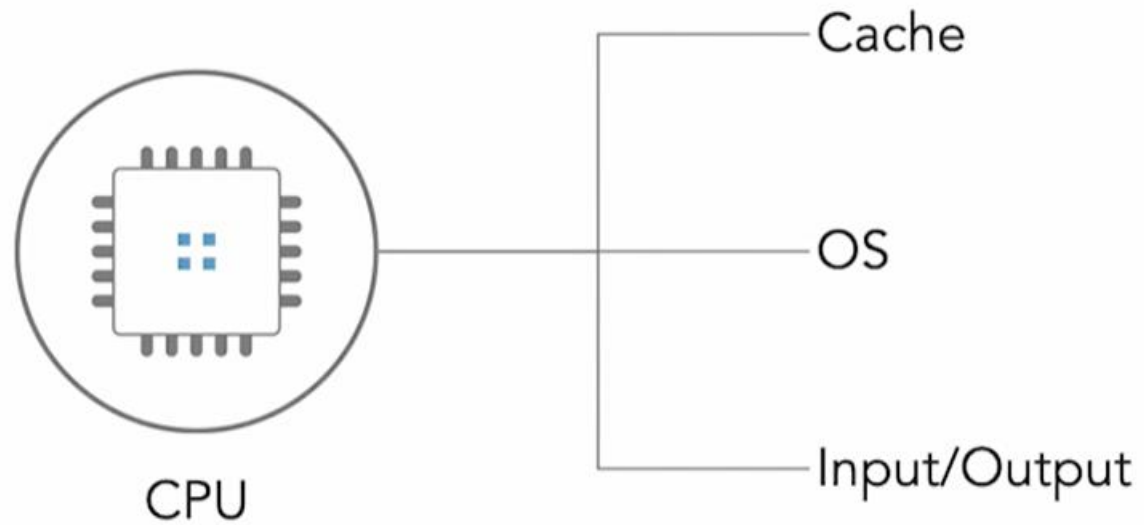
Edge computing networking chain



Edge computing storage



Edge computing compute



Compute

- Within the cloud
- Shared
- Function around a particular utility

Edge management

Abstraction

Cloud 1

Cloud 2

Cloud 3

Edge
Device

Edge
Device

Edge
Device

Edge
Device

Edge
Device

Edge
Device

Edge
Device

Edge
Device

Edge
Device

Edge
Device

Edge
Device

Edge management includes

- Security management
- Governance
- Service
- Resource
- Data management
- Network management
- Configuration management
- Storage management

Edge computing requirement

Security	Operations	Governance	Management	Data Management
• Sensors • Device • Back-end cloud-based systems	• Consistent • Automated • Visibility	• Use of services • Use of AI systems • Not the same as security	• Coexists with operations • Long-term operation • Deal with updates	• Data stored in different places • Cloud based • Centralized

Other consideration

- AI and machine learning
- Hardware management
- Network management
- Device-to-cloud management

Edge computing: security

Secure everything, moving through the network or sitting on a physical device.

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11001110100011001
00101010110000100
01010010101000011
00001011111100010



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00101010110000100
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00001011111100010



Detect Threat

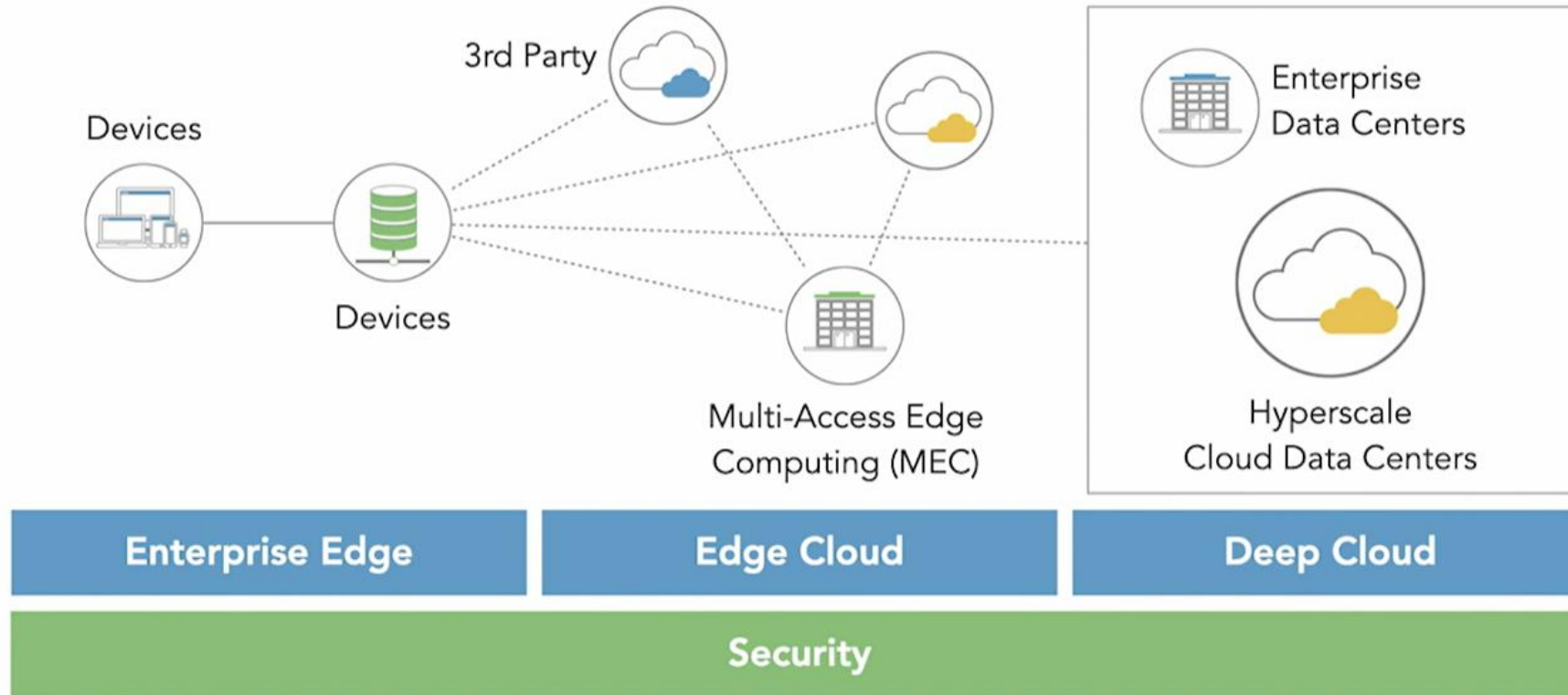


Proactive Threat Detection

Detect a threat before it happens.

- Automation Automated patching link that to operations

Security runs end to end



Edge governance



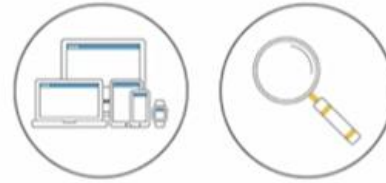
Users



Applications

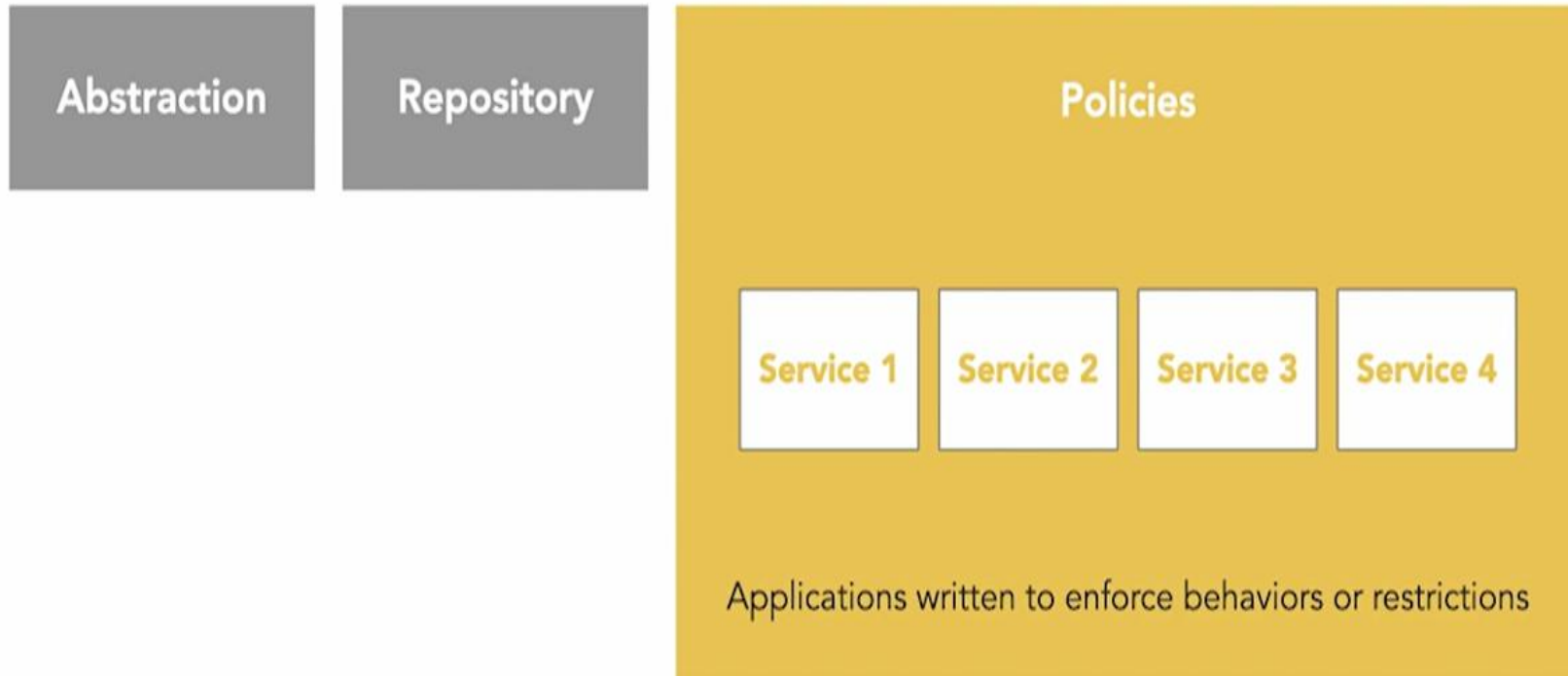


Policies



Edge Devices
and Data

Service governance

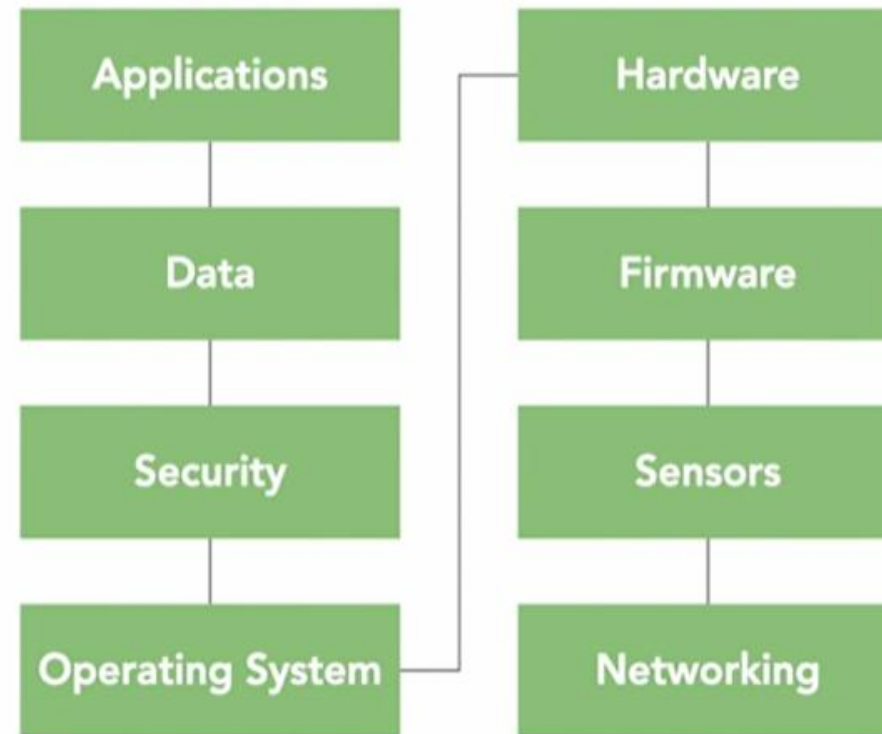


Security and governance

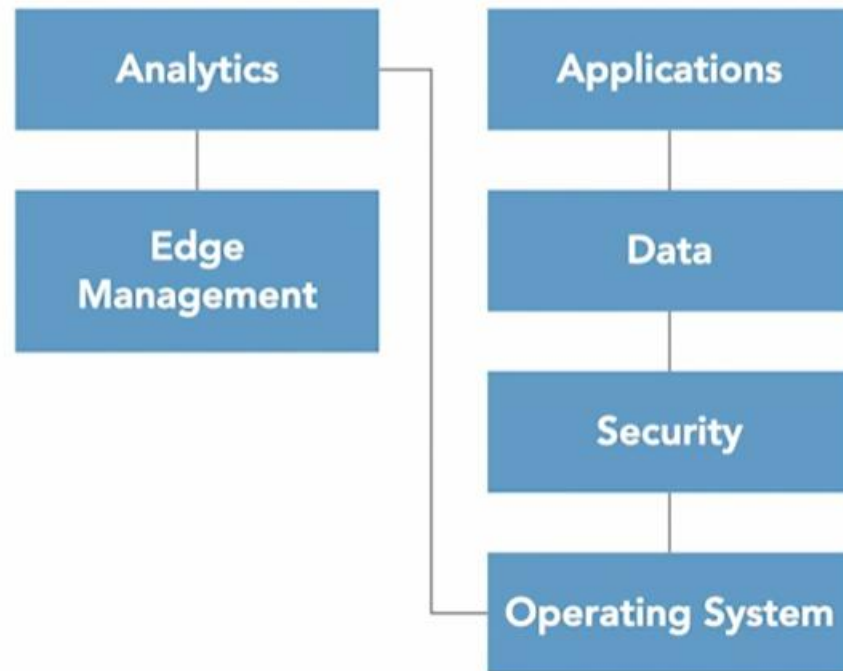


Edge configuration management

Device side



Cloud side

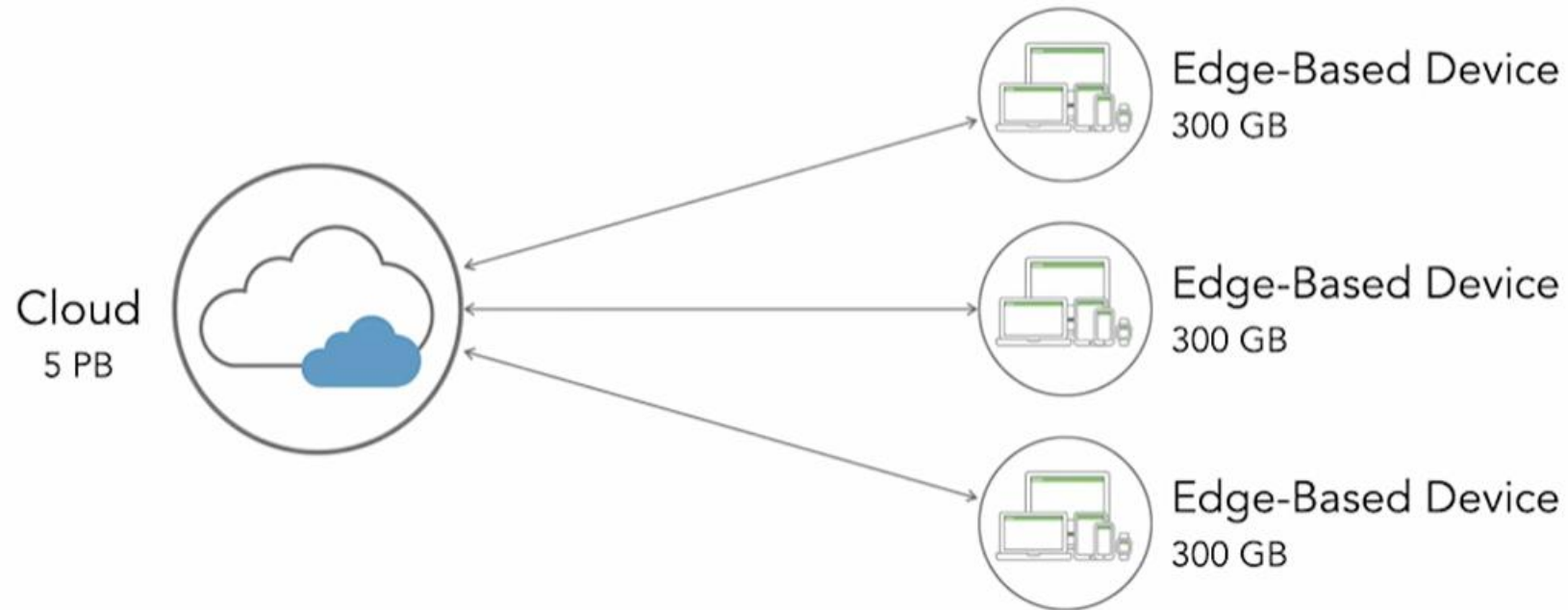


Edge computing data management

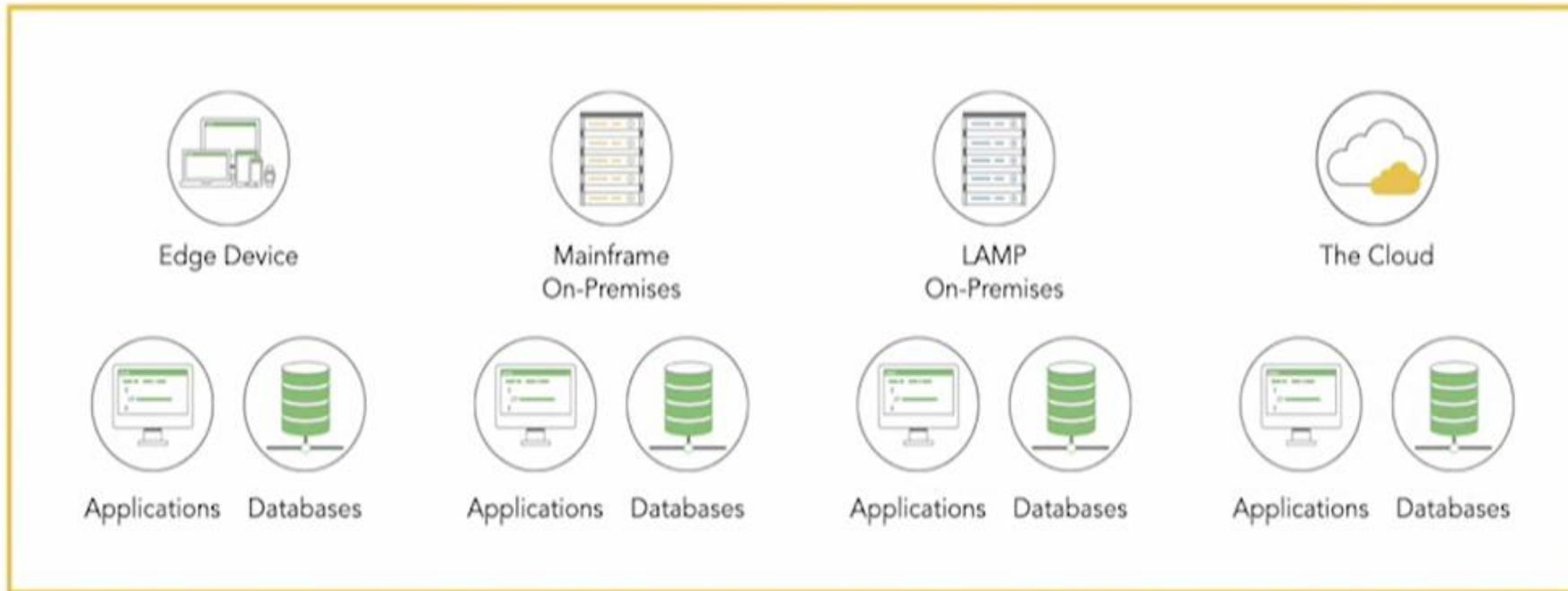
- Storage
 - Metadata
 - Data security
 - Business continuity and disaster recovery (BC/DR)
 - Change management
- Storage and data synchronization should be built into your edge computing design

Tiered or no replication

Replicate device to cloud.



Edge computing operation and monitoring



Edge operation

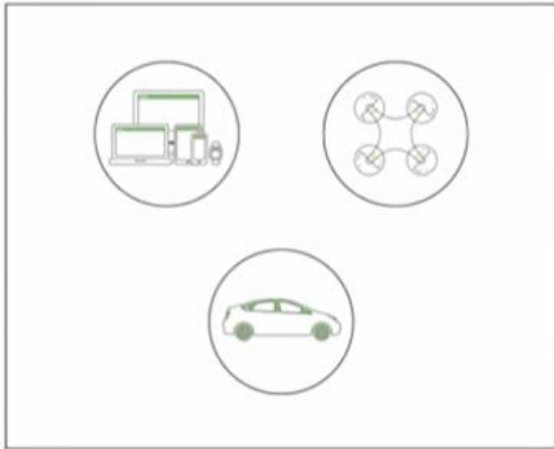
- BC/DR
- SecOps
- Network Ops
- Management
- Monitoring
- Special purpose
- Applications
- Knowledge base

The process

- 1 Understanding your edge computing and operations requirements
- 2 Creating your edge computing operations plan
- 3 Selecting the right monitoring and operations tools
- 4 Defining monitoring operations patterns for your enterprise

Edge computing

Edge computing and cloud computing typically work well together.

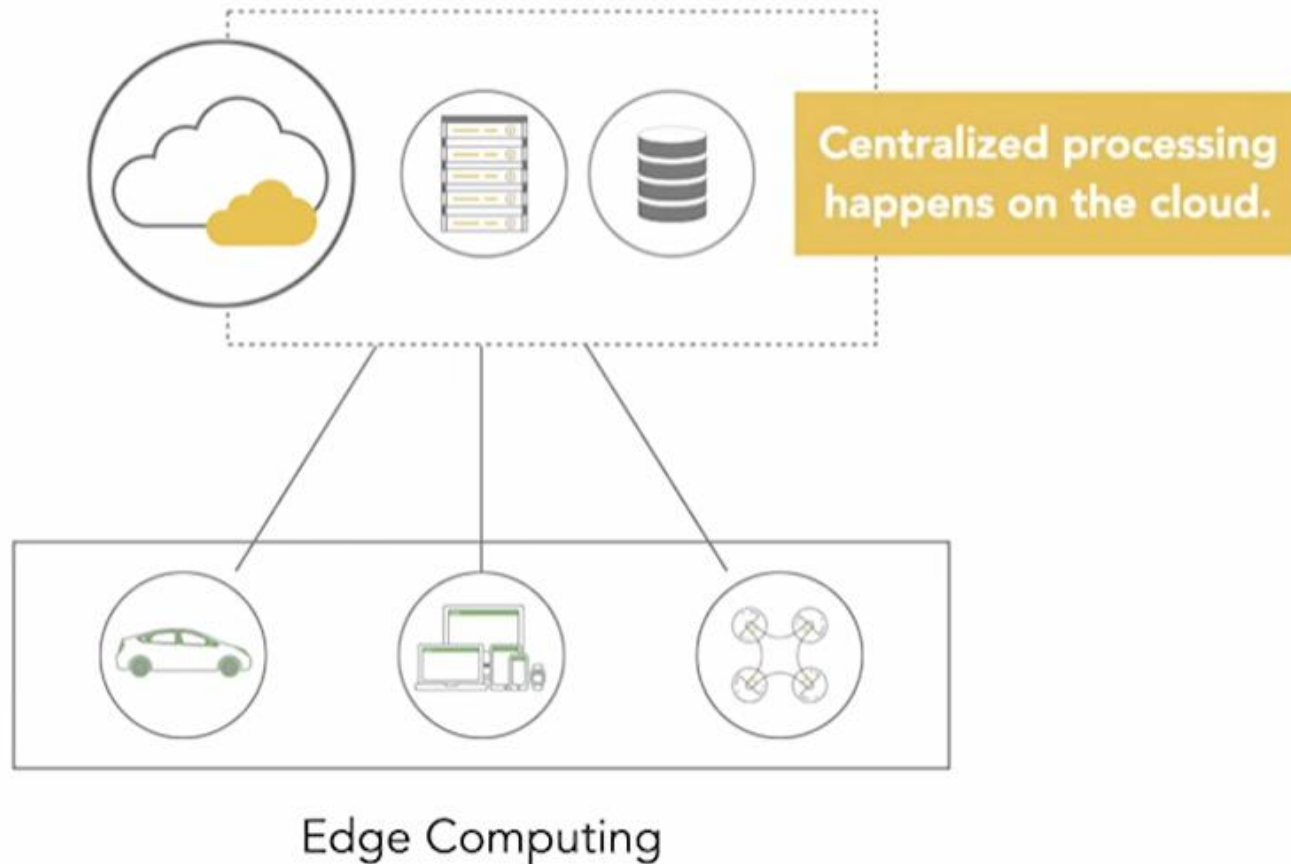


Edge Computing



Cloud Computing

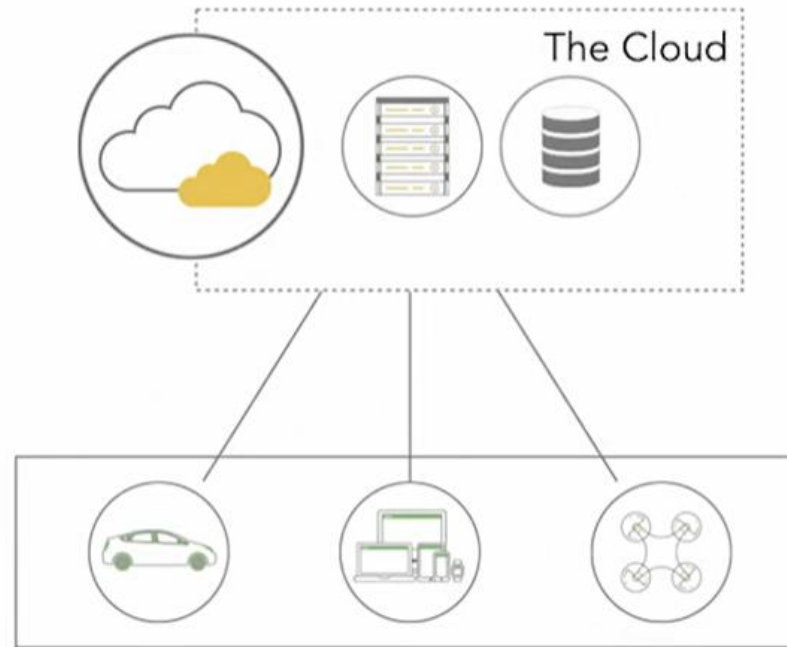
Edge computing and cloud computing work together to create a proper edge computing architecture.



Many edge systems start in the cloud and move to the edge.



Cloud Computing

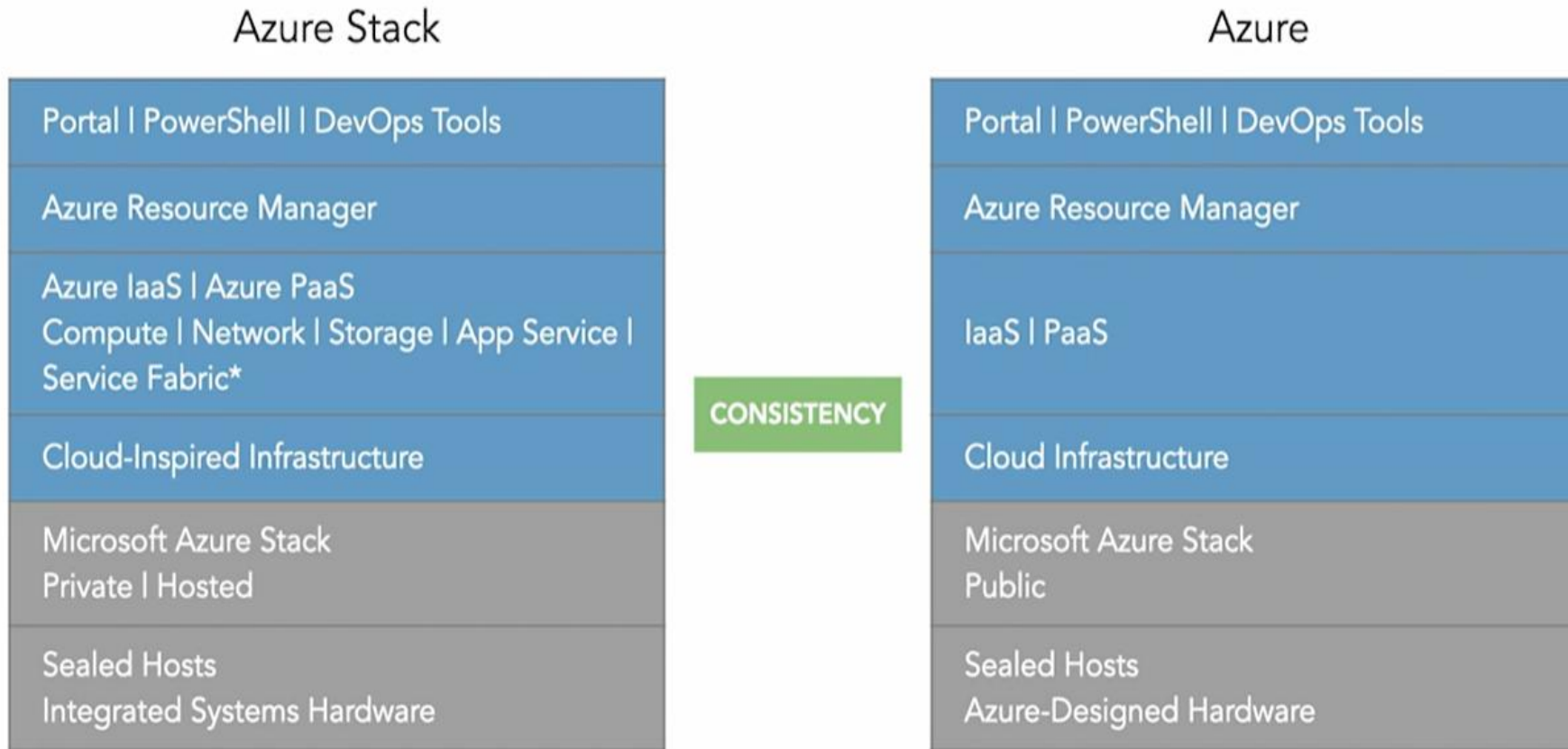


Edge Computing

Types of edge services

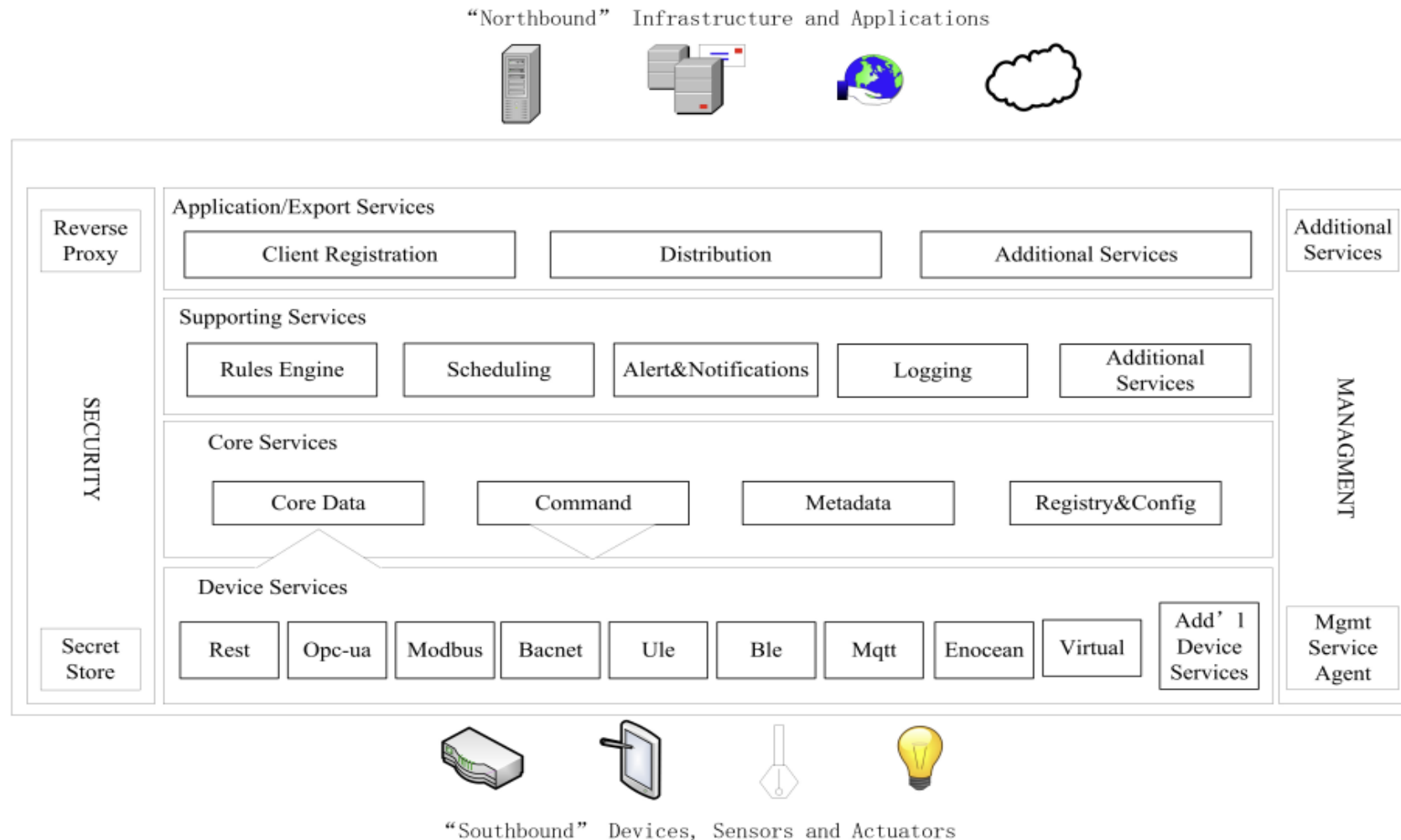
- Micro clouds – AWS Outposts and Azure Stack
- Edge system management – AWS IoT SiteWise Edge
- Security
- Governance
- More complexity moving to the edge

Azure stack and azure



Azure Stack is a portfolio of products that extend Azure services and capabilities to your environment of choice

EdgeX foundry architecture

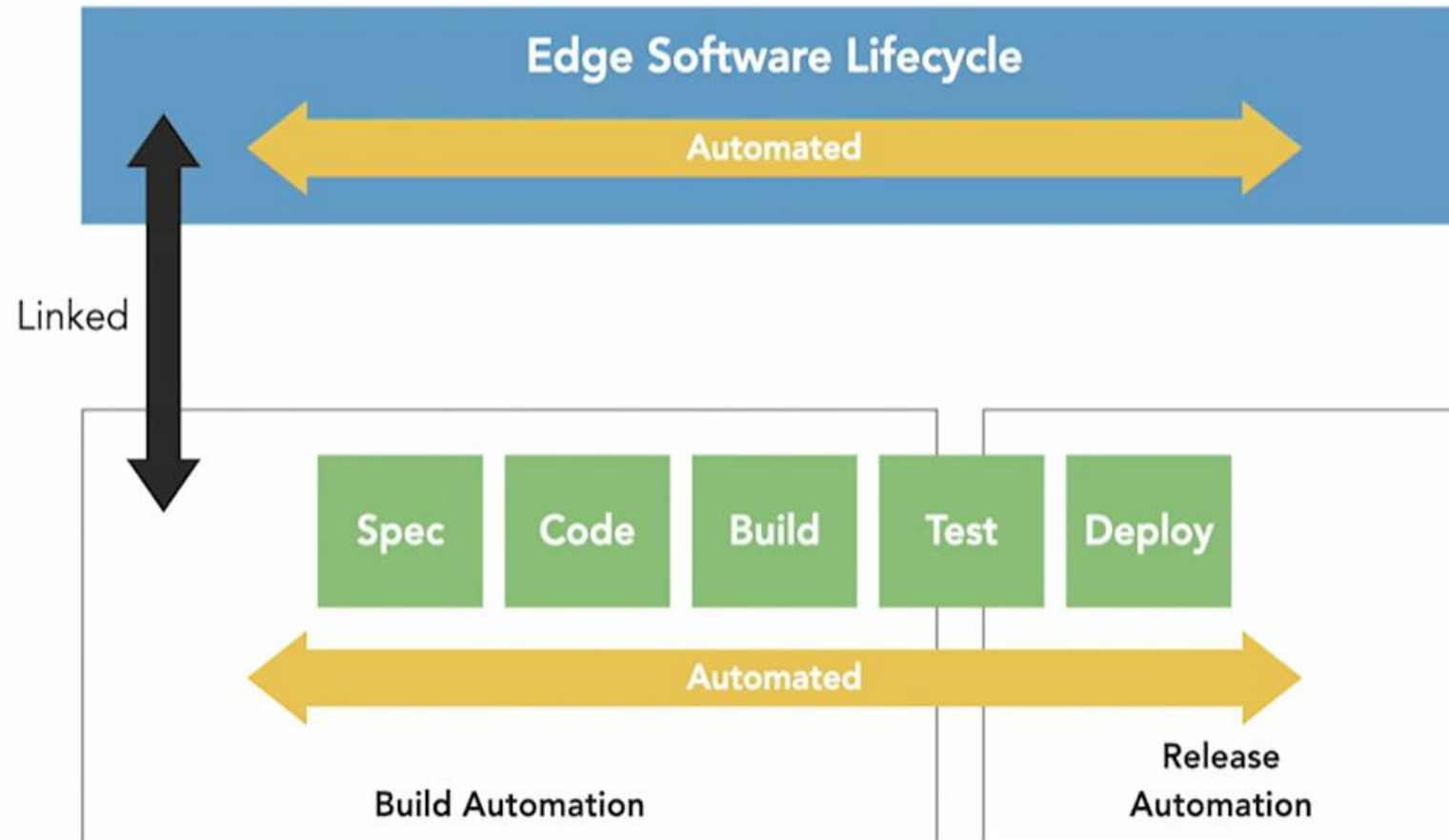


EdgeX Foundry is a neutral open source project hosted by the Linux foundation and is a universal open framework for computing on the edge of the Internet of Things. Hosted on a reference software platform that is completely independent of hardware and operating systems, the framework enables a plug-and-play component ecosystem.

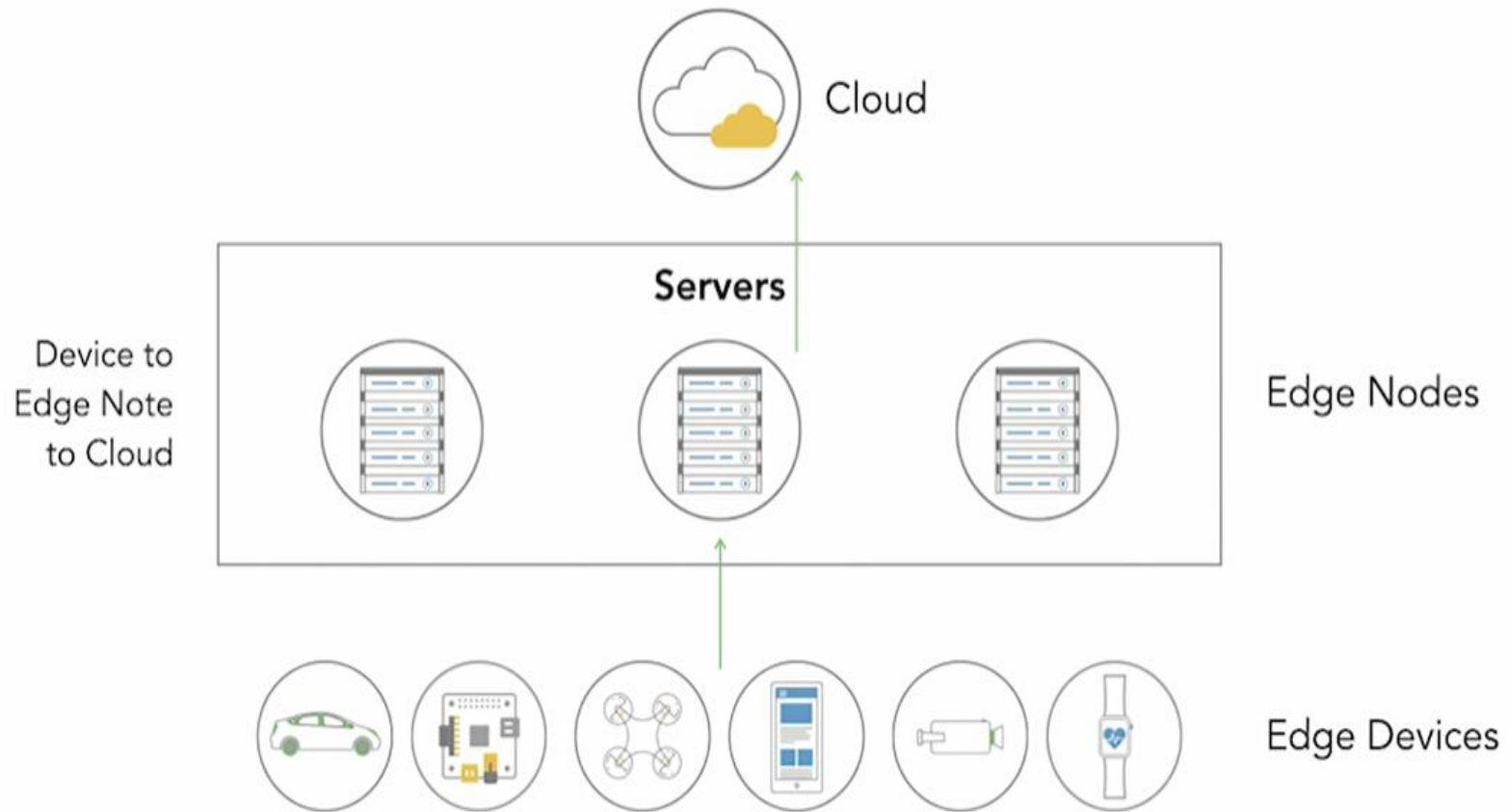
Development to deployment



Edge computing software development

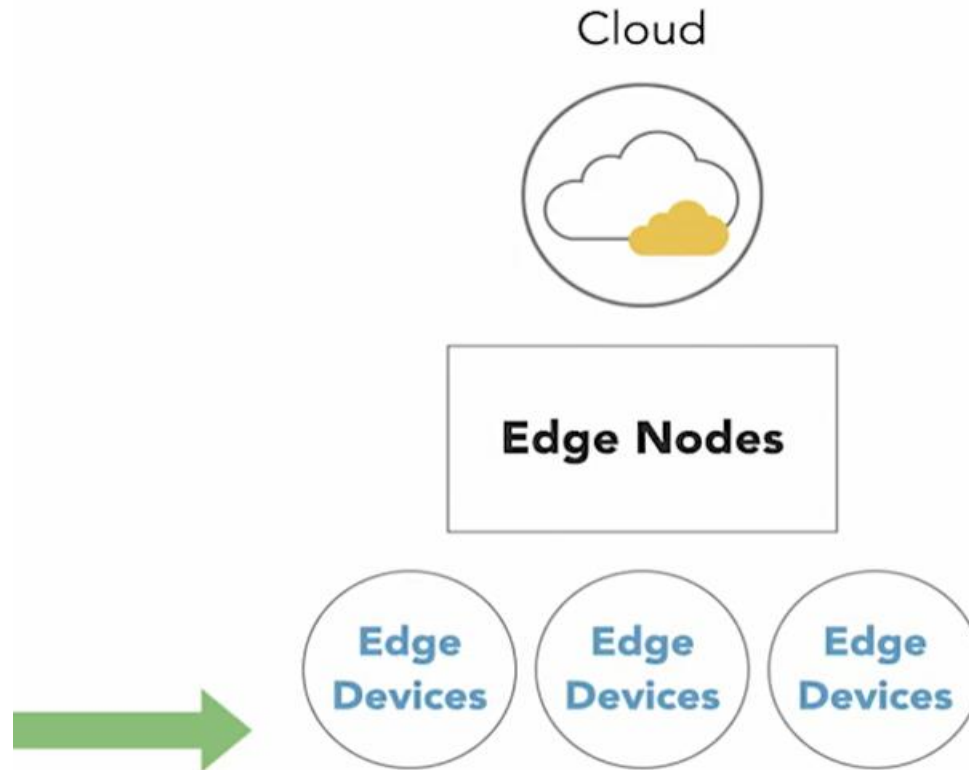


Edge computing



Edge development process

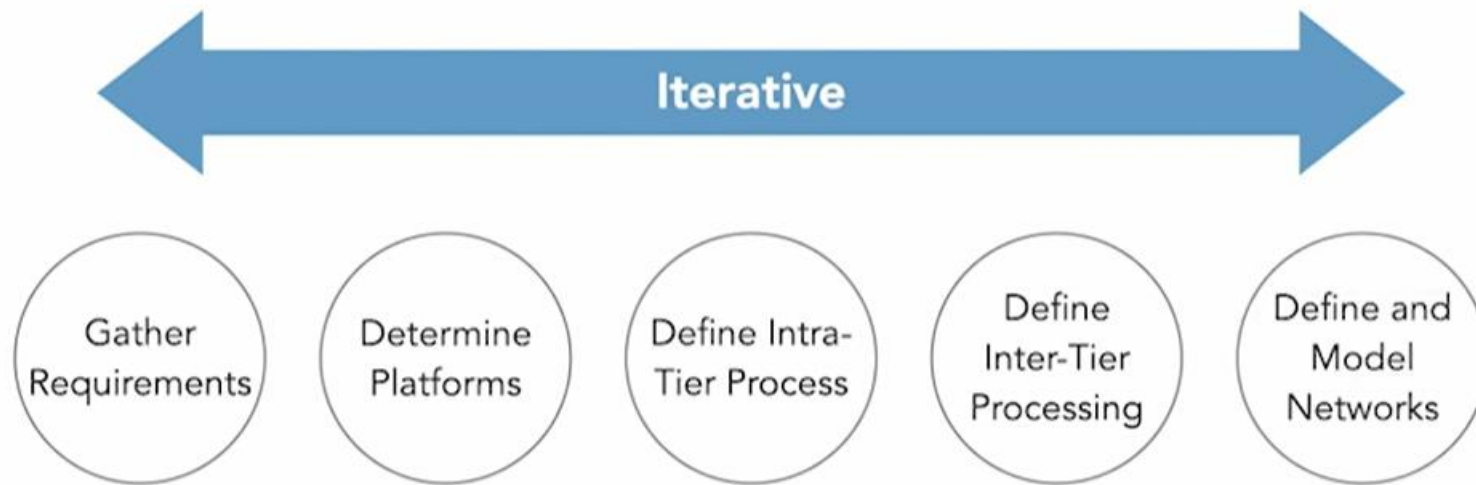
- Define the application
- Define data
- Build the edge system
- Test the system
- Deploy the edge system



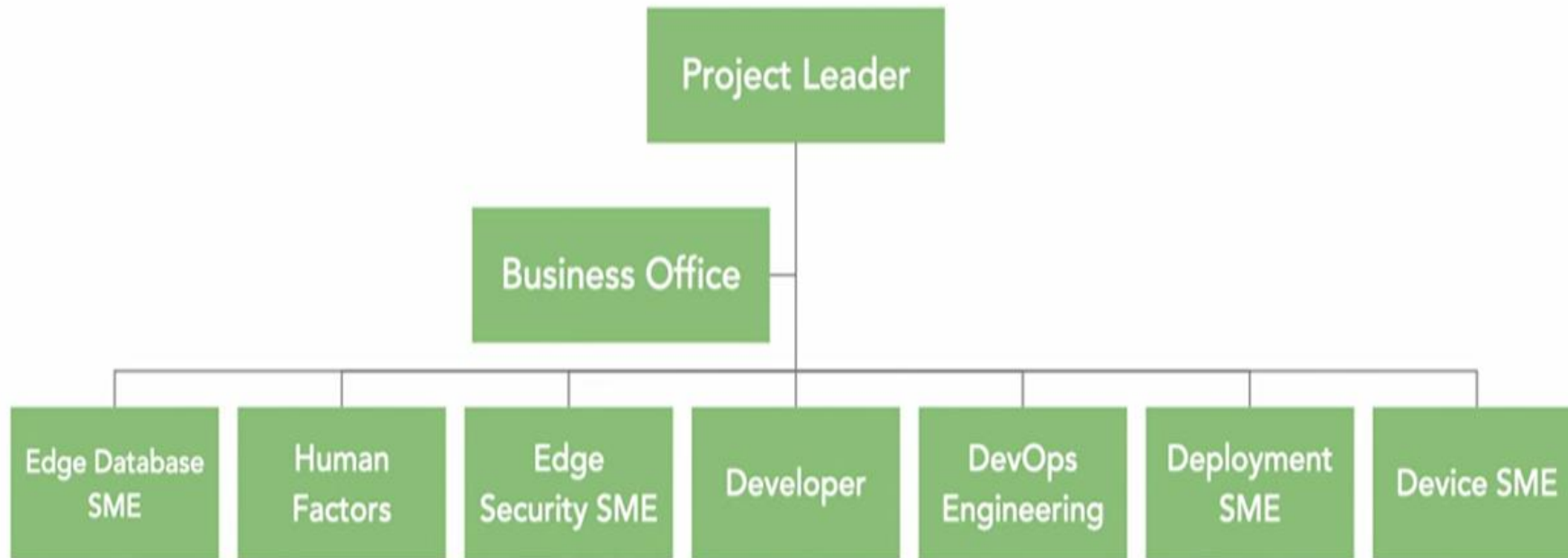
New in edge computing

- Tiered distributed computing system
- Distributed and heterogenous data
- Complex security and governance
- Complex operations

Meta design process



The edge Team



Applying dev Ops to Edge computing development

Spec	Design
Code	Development
Build	Integration
Test	Testing
Deploy	Deployment
CloudOps and EdgeOps	Move to Ops



Edge computing DevOps tools

DevOps Domains and Capabilities

	Work and Issue Tracking and Reporting	Requirements Management	Test-Driven Development	Application Portability	Continuous Integration	Automated Testing	Continuous Deployment	Dynamic Environments	Continuous Monitoring	SOURCE CODE MANAGEMENT	ARTIFACT REPOSITORY	CONFIGURATION MANAGEMENT
	JIRA RALLY LeanKit VERSIONONE TeamForge	Confluence JIRA blueprint HP Agile Manager	JUnit Selenium Cucumber thucydides	docker Orkt	Bamboo Jenkins go TeamCity	Selenium SoapUI Cucumber SpecFlow	Bamboo Octopus Deploy go	Amazon AWS Google Cloud Platform VMware	Iqant Nagios Elastic.io Logstash New Relic	Stash GITLAB GitHub Enterprise SUBVERSION ali ClearCase	artifactory Amazon S3 Nexus HP ALM urban(code)	puppet labs CHEF ANSIBLE VAGRANT
MICROSOFT	Team Foundation Server	Team Foundation Server	Microsoft Test Manager	Visual Studio	Team Foundation Server	Microsoft Test Manager	Team Foundation Server	Azure Hyper-V System Center	System Center			
IBM	IBM Rational	IBM Rational	IBM Rational		urban(code) uBuild	IBM Rational Performance Tester	urban(code) uRelease uDeploy	IBM Cloud™ SOFTLAYER™	HP Tivoli Suite			

Deployment phase



Edge Application deployment tool

- Version control
- Code review
- Continuous integration (CI)
- Configuration management
- Release automation
- Infrastructure monitoring

Building edge ops process

Understand Your Requirements

- Computing
- Monitoring
- Operations
- Long-term deployment

Select Tools

- Configuration
- Automation
- Monitoring
- Abstraction

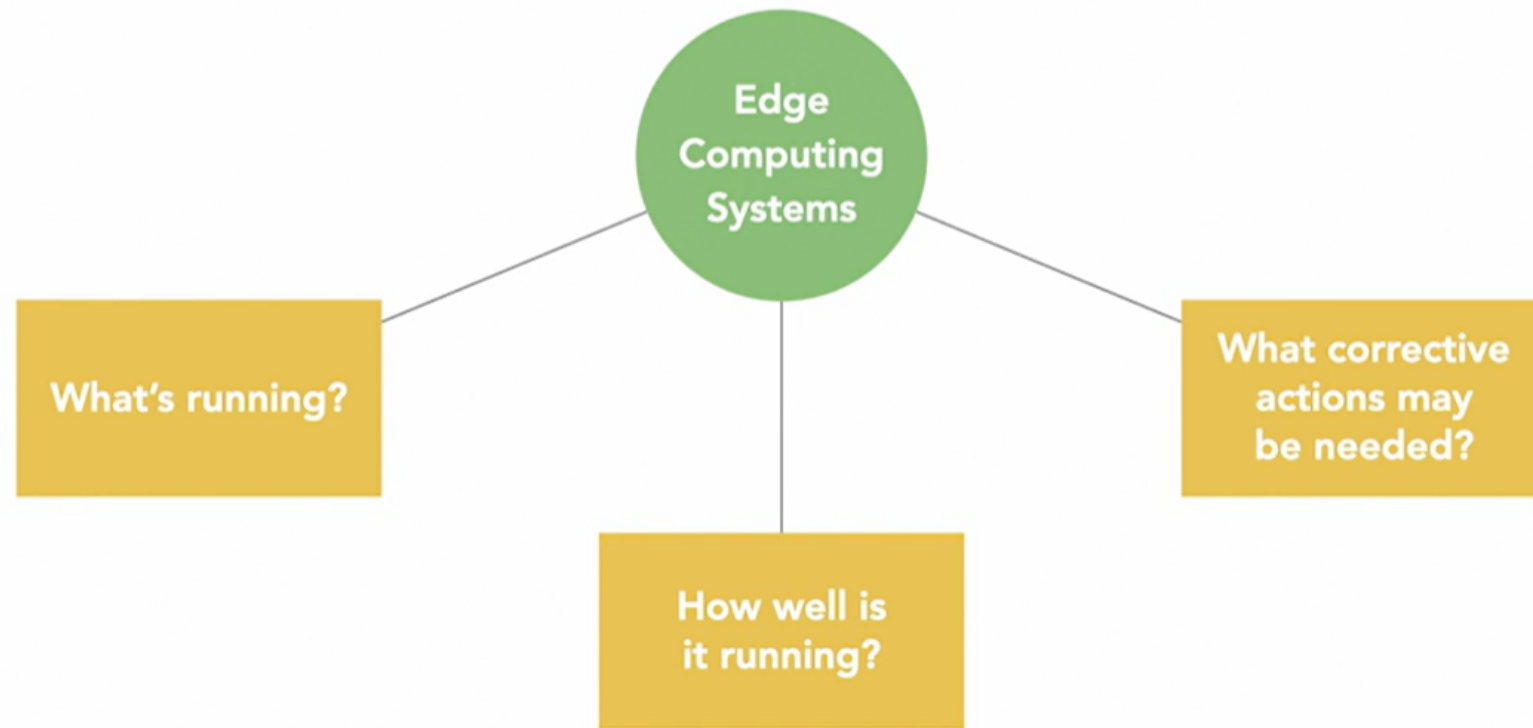
Create a Plan

- How the system will run
- Ability to make changes

Define Your Enterprise

- Localized operations
- Specific requirements
- Customized security
- Business privacy needs

The General idea



Edge computing monitoring tools

- Database
- Applications
- Security systems
- Performance
- Network
- Native cloud service
- Edge node or device
- Custom monitoring

Future of Edge



Building Edge Ops process



5G



Better Complexity
Management



Better Cross-Tier
Portability



Database
Advancement



Standards for
Devices and Sensors



More Deployments to House, Car, Humans

AI can be everywhere in Tiered



Public Cloud

AI engines
Mass training data
Analytics



Edge Server

AI engines
Most training data
Analytics

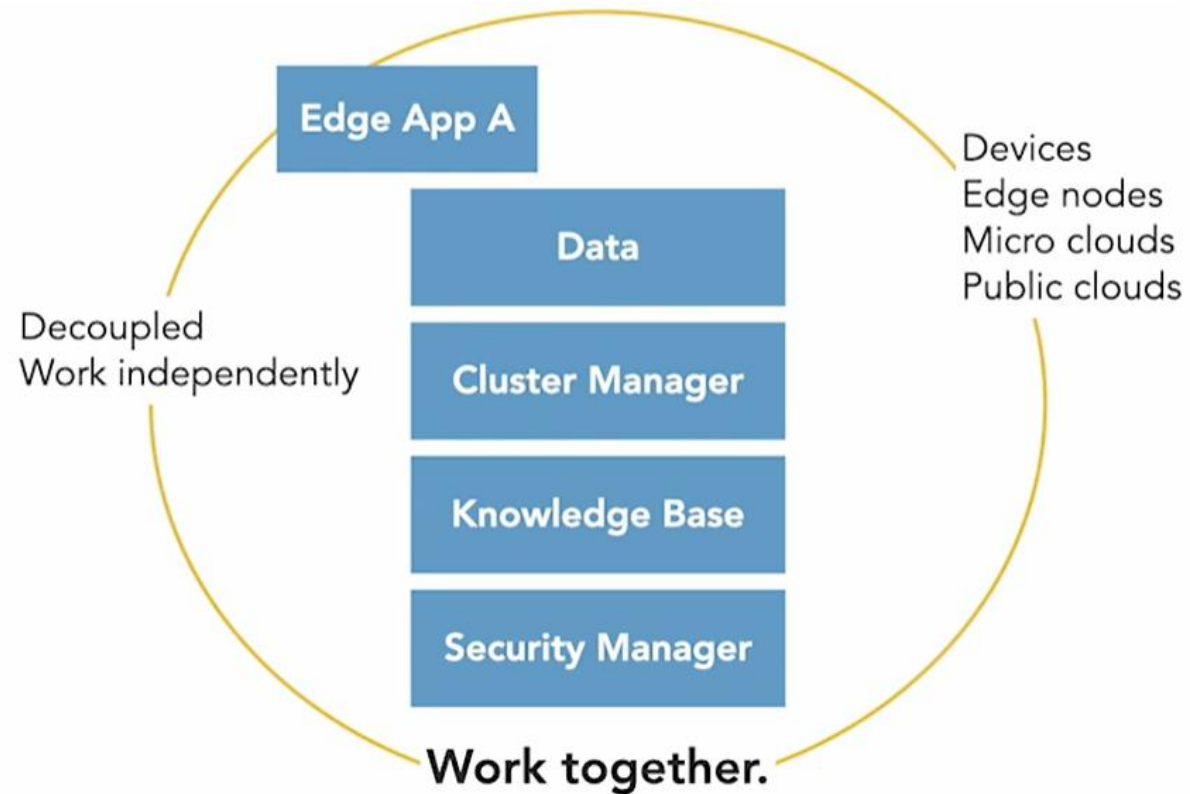


AI engines
Some training data
Instant engagement

Meeting common requirements

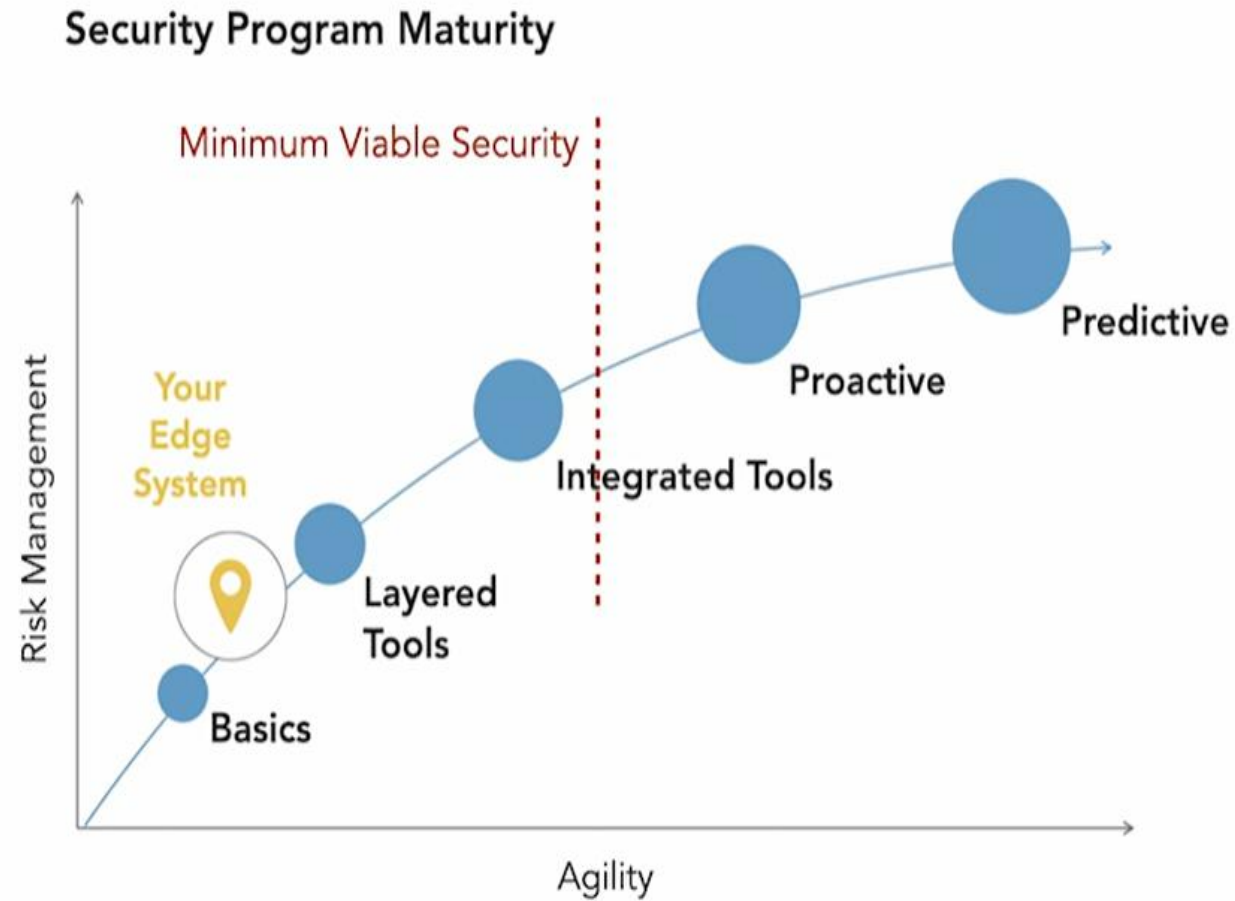
- When patterns can be leveraged
- When dynamic data is available
- When automation is a benefit
- When the data is available
- Anytime applications benefit from gather and leveraging knowledge

Federated Architecture



Could run on its own:
Device
Server
Process in the cloud

Federated Architecture



Types of edge computing security tools

- Governance
- Identity
- Authentication
- Authorization
- Audit
- Encryption
- Compliance

Future Edge based Application

- Microrobotics
- Reduce carbon emissions
- Automated portability
- New architectures
- Standards emerge

Your role

- Look at public cloud edge services
- Experiment with devices
- Build simple applications
- Establish knowledge of architecture and design
- Play with some tools

References

- ▶ 1. Book - Internet of Things for Architects
- ▶ 2. A Survey on Mobile Edge Computing - Yuyi Mao, Changsheng You, Jun Zhang, Kaibin Huang, and Khaled B. Letaief 2017
- ▶ 3. Deep Learning with Edge Computing: A Review Jiasi Chen, Xukan Ran 2019
- ▶ 4. Edge Computing: Vision and Challenges Weisong Shi, Jie Cao, Quan Zhang, Youhuizi Li and Lanyu Xu 2016

Question



Thank you

